SOWNDAPPAN S

LLM Engineer · Agentic AI Systems · RAG Architect

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PROFESSIONAL SUMMARY

LLM Engineer with production experience architecting agentic AI systems, RAG pipelines, and fine-tuned language models. Built a self-extending LangGraph agent framework supporting 50+ runtime-generated tools with 100% execution accuracy. Designed a production RAG system benchmarked across 4 chunking strategies, improving retrieval coverage to 68% at sub-second latency. Fine-tuned Qwen2.5-1.5B with LoRA and 4-bit quantization at ~99% parameter efficiency, with a peer-reviewed publication in applied CV as supporting research depth.

CORE COMPETENCIES

LLM & Agentic AI: LangChain, LangGraph, Agentic Orchestration, Dynamic Tool Generation, Prompt Engineering, RAG, Structured Outputs, Gemini API, Groq API

Fine-Tuning & Deep Learning: PyTorch, HuggingFace Transformers, PEFT, TRL, LoRA/QLoRA, Transfer Learning

Backend & APIs: FastAPI, Flask, REST API Design, Pydantic, PostgreSQL, MongoDB, ChromaDB

Cloud & DevOps: AWS Lambda, CloudWatch, Git, GitHub, Linux

Applied ML / CV: YOLOv8, OpenCV, Object Detection, Monocular Depth Estimation

PROFESSIONAL EXPERIENCE

Data Science Intern (AI/LLM Focus) — Logesys Solutions India Pvt. Ltd
Sep 2025 – Present | Bengaluru, India*

- Engineered 10+ production AI APIs serving 500+ requests/day, powering assessment generation, personalized recommendations, and learning analytics for 10,000+ learner interactions.
- Built LLM-based question generation and automated explanation pipelines using LangChain, LangGraph, and Gemini API, generating 1,000+ assessment questions and reducing manual authoring effort by 40%.
- Designed prompt and structured-output strategies for reliable generation quality across adaptive assessment and adaptive learning features.
- Deployed and monitored services on AWS Lambda + CloudWatch, closing the loop between production behavior and prompt/model iteration.

KEY PROJECTS

AdaptBot — Dynamic LLM Agent Orchestration Framework · *Python, LangGraph, LangChain, Pydantic, Groq API* [GitHub]

- Architected a self-extending AI agent capable of creating and executing 50+ runtime-generated tools without graph recompilation, using a static 4-node LangGraph pipeline with dynamic tool loading.
- Benchmarked against GPT-OSS-120B: 93.2% intent classification, 100% parameter extraction + tool execution success; implemented structured Pydantic extraction with sub-second tool registration via a runtime registry.

Production RAG System — Enterprise Document Intelligence · *Python, FastAPI, ChromaDB, Sentence Transformers, Groq API* [GitHub]

- Benchmarked 4 RAG chunking strategies (fixed, sentence, paragraph, recursive) across 100+ evaluation questions and 1,000+ indexed chunks, measuring retrieval coverage, answer accuracy, latency, and throughput.
- Demonstrated recursive chunking improved retrieval coverage from 58% to 68% while maintaining sub-second latency, reducing context fragmentation on long-form documents.
- Built a FastAPI + ChromaDB evaluation harness with independent vector indexes for fair side-by-side strategy comparison; supports PDF, DOCX, and TXT ingestion.

Commit Message Generation — LoRA Fine-Tuning · *Python, PyTorch, Transformers, PEFT, TRL* [GitHub]

- Fine-tuned Qwen2.5-1.5B-Instruct on 10,000 GitHub commit messages using LoRA and 4-bit NF4 quantization to generate professional commit summaries from informal inputs.
- Reduced trainable parameters by ~99% vs full fine-tuning, enabling single-T4-GPU training via an SFTTrainer instruction-tuning pipeline.

OPEN-SOURCE CONTRIBUTIONS

- **DeepSpeed (Microsoft)** — 3 merged PRs improving quantization config parsing, optimizer regression testing, and training-infra reliability.
- **LangChain** — Added schema validation checks and regression tests, improving structured-output API reliability.
- **Caura AI** — Improved MCP search validation; built fault-tolerant memory enrichment with automatic fallback and test coverage.
- **Recursion Pharma (GFlowNet)** — Optimized graph-processing via lightweight pre-filtering, cutting unnecessary isomorphism checks by 3–7%.

PUBLICATION

Monocular Object Distance Estimation using Calibration-Based Perspective Mapping and Detection Integration

IRJET, Vol. 13, Issue 04, Apr 2026 · 98% interval accuracy, 0.36 m MAE, O(1) CPU-only real-time inference

ACHIEVEMENTS

Top 5, Alliance University CodeSangram 30-Hour Hackathon · Best Student Project Award, Computer Society of India.

EDUCATION

B.Tech in Information Technology **CGPA: 8.77 / 10**

Knowledge Institute of Technology, Salem · Nov 2022 – May 2026